

Math 115A Spring 2026 Discussion 17 Handout

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1 Introduction

Today we'll solve some problems about inner product spaces.

2 Inner Products

As a reminder:

Definition: Let V be a vector space over $\mathbb{R}$ or $\mathbb{C}$. An inner product on V is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow F$ satisfying:

1. For each x , $\langle \cdot, x \rangle : V \rightarrow F$ is linear
2. For each $x \in V$, $\langle x, x \rangle \in \mathbb{R}$, and $\langle x, x \rangle > 0$ for $x \neq 0$ (and $\langle 0, 0 \rangle = 0$).
3. $\langle y, x \rangle = \overline{\langle x, y \rangle}$ for all $x, y \in V$

Over $\mathbb{R}$, the main example is the dot product in $\mathbb{R}^n$, with which I'm sure you're all familiar. Perhaps less famous is the complex dot product in $\mathbb{C}^n$: for $v = (v_1, \dots, v_n), w = (w_1, \dots, w_n)$, we have $v \cdot w = \sum_i v_i \overline{w_i}$.

The real dot product has a well-known geometric interpretation: it measures the cosine of the angle between two vectors of known length. The complex dot product is a bit more obscure - the justifications of it I've seen go a bit too deep for Math 115A - so for now you should think of it as a formal algebraic device.

In either case, dot products serve to give us a notion of angles. But to get some practice with the algebraic definition (without relying on too much geometric intuition), here's an example that *isn't* about angles (at least not on its face...):

Exercise: Let $V = L^2([0, 1])$ denote the set of Riemann-integrable functions $f : [0, 1] \rightarrow \mathbb{R}$ such that $\int_0^1 |f(x)|^2 dx < \infty$. This turns out to be an $\mathbb{R}$ -vector space. (You can take my word that this is closed under addition - the proof is not too bad, but is beyond the scope of this class). Show that the function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ given by $\langle f, g \rangle = \int_0^1 f(x)g(x)dx$ is an inner product.

Remark: If one considers instead functions $f : \mathbb{R} \rightarrow \mathbb{C}$, we have a notion of "square integrable:" f is square integrable over, say, $[0, 1]$, if $\int_0^1 |f(x)|^2 dx < \infty$ (where here $|f(x)|$ is the complex absolute value: $|a + bi| = \sqrt{a^2 + b^2}$). The

space W of complex square-integrable functions has an inner product $\langle \cdot, \cdot \rangle_{\mathbb{C}} : W \times W \rightarrow \mathbb{C}$ given by

$$\langle f, g \rangle_{\mathbb{C}} = \int_0^1 \overline{f(x)}g(x)dx$$

which, if you'd like, you can verify.

A vector space with a specified inner product is called an *inner product space*. Of course, every inner product space has a basis.

Exercise: Let V be a vector space over a field F , let $(W, \langle \cdot, \cdot \rangle_W)$ be an inner product space over F , and let $f : V \rightarrow W$ be an *injective* linear map. Define $\langle \cdot, \cdot \rangle_f : V \times V \rightarrow F$ by

$$\langle v_1, v_2 \rangle_f = \langle f(v_1), f(v_2) \rangle_W$$

Show that this is an inner product on V . Why is this not an inner product if f is not injective?

Such an inner product is often called a *pullback inner product*. (This is part of a more general “pullback” phenomenon in mathematics, where given two objects $X \rightarrow Y$, a map $f : X \rightarrow Y$, and some additional structure on Y , we can construct a similar structure on X). In a moment, we'll see how powerful this technique can be.

Definition: Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Let $\mathcal{B}$ be a basis for V .

1. We say that $\mathcal{B}$ is *orthogonal* if for all $v, w \in \mathcal{B}$ such that $v \neq w$, we have $\langle v, w \rangle = 0$.
2. We say that $\mathcal{B}$ is *normal* if for all $v \in \mathcal{B}$, we have $\langle v, v \rangle = 1$.
3. We say that $\mathcal{B}$ is *orthonormal* if it is both orthogonal and normal.

Exercise: Consider $\mathbb{R}^n$ with the standard inner product. Show that the standard basis for $\mathbb{R}^n$ is orthonormal.

Exercise:

- (a) Let V be a vector space and $(W, \langle \cdot, \cdot \rangle)$ be an inner product space. Let $f : V \rightarrow W$ be an *isomorphism* - in particular, f is injective, so we can construct the pullback inner product $\langle \cdot, \cdot \rangle_f : V \times V \rightarrow \mathbb{R}$. If $\mathcal{B} = \{w_1, \dots, w_n\}$ is an orthonormal basis for $(W, \langle \cdot, \cdot \rangle_W)$, show that $f^{-1}\mathcal{B} = \{f^{-1}w_1, \dots, f^{-1}w_n\}$ is an orthonormal basis for $(V, \langle \cdot, \cdot \rangle_f)$. (**Hint:** Start by making a list of all the things you need to check!)
- (b) Let V be a finite dimensional vector space and let $\mathcal{B} = \{v_1, \dots, v_n\}$ be a basis for V . Show that there is an inner product $\langle \cdot, \cdot \rangle_V : V \times V \rightarrow \mathbb{R}$ with respect to which $\mathcal{B}$ is orthonormal. (Hint: use part (a))

The above exercise shows that every finite-dimensional vector space can be viewed as an inner product space in many different ways.¹ What an inner product gets you is a preference for certain bases over others - that is, an inner product endows a vector space with geometry (lengths and angles) and thus prioritizes bases with good geometric properties.

¹For infinite-dimensional vector spaces the situation is a bit more complicated - you can always construct an inner product, but it might not cooperate with other geometric structure. For example, there are infinite-dimensional vector spaces V with natural notions of length which can be shown not to come from inner products, i.e. there is no inner product such that $\|x\|^2 = \langle x, x \rangle$ for all x .